

Chasing Wild Horses: Evaluating Music Content Analysis with Irrelevant

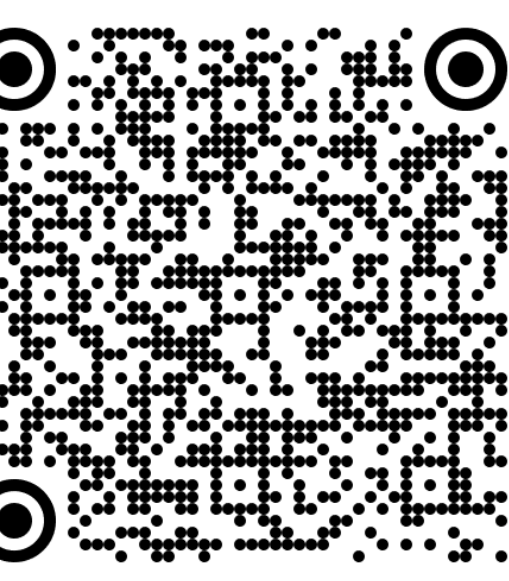
Konrad Swierczek¹, Michael Schutz²

swierckj@mcmaster.ca

schutz@mcmaster.ca

¹ Department of Psychology, Neuroscience & Behaviour, McMaster University

² School of the Arts, McMaster University



Background

- Evaluation is central to advancing Music Information Retrieval (Downie, 2004; Sturm 2016)
- Different versions of the same composition result in inconsistent extractions of structural features like mode and onsets (Swierczek & Schutz, under review)

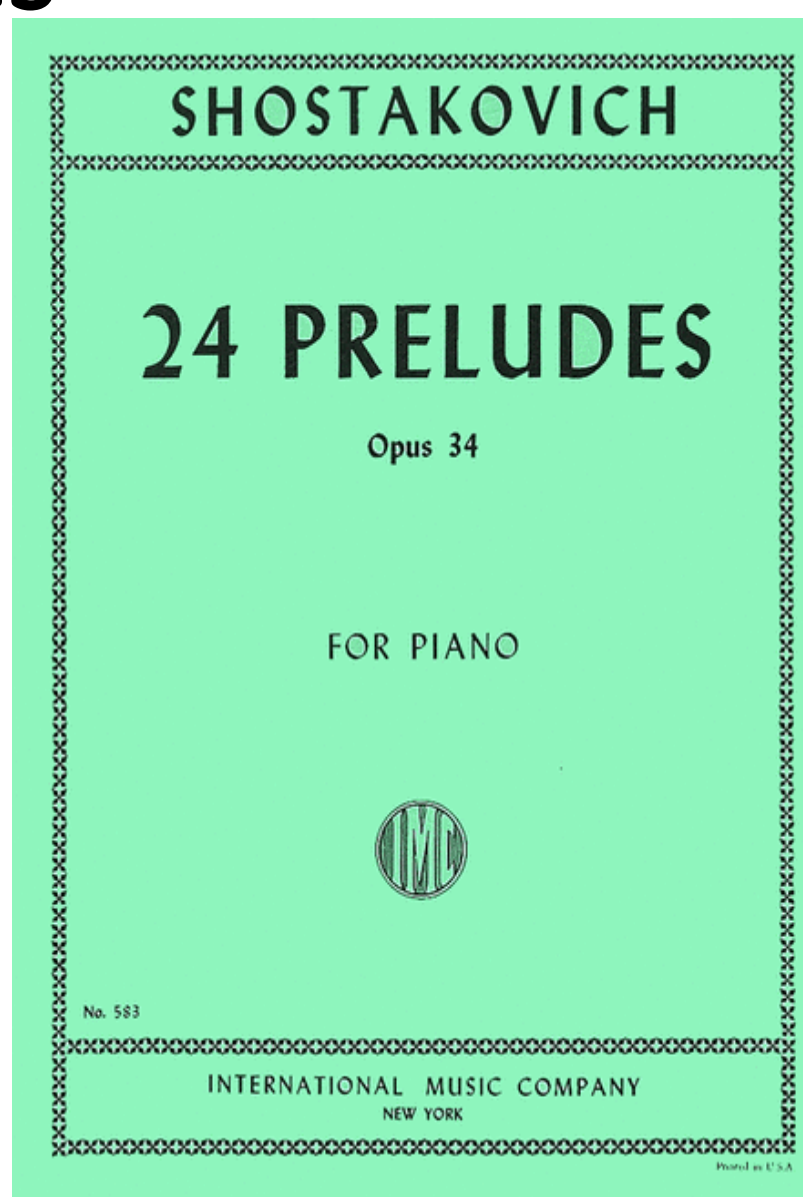
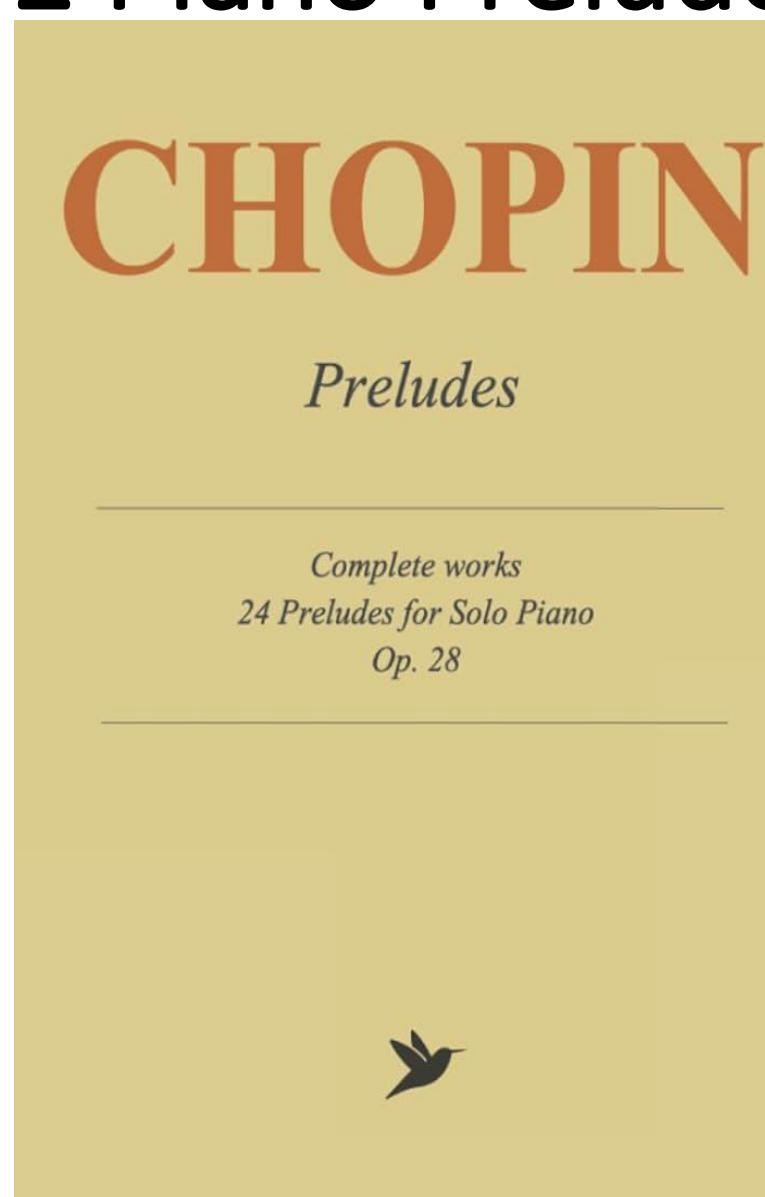
How do specific musical and audio properties influence feature extraction?

- We synthesize audio from MIDI to examine the individual influence of manipulations on a variety of features and analysis tools

Method



72 Piano Preludes



Summary

- We propose a method for evaluating MIR features that does not rely on ground truth
- We examine five manipulations that may influence the extraction of features
- Onset detection is influenced by MIDI manipulations, all features are robust to quality.
- **Applications:** evaluation, algorithm selection, parameter optimization

Results

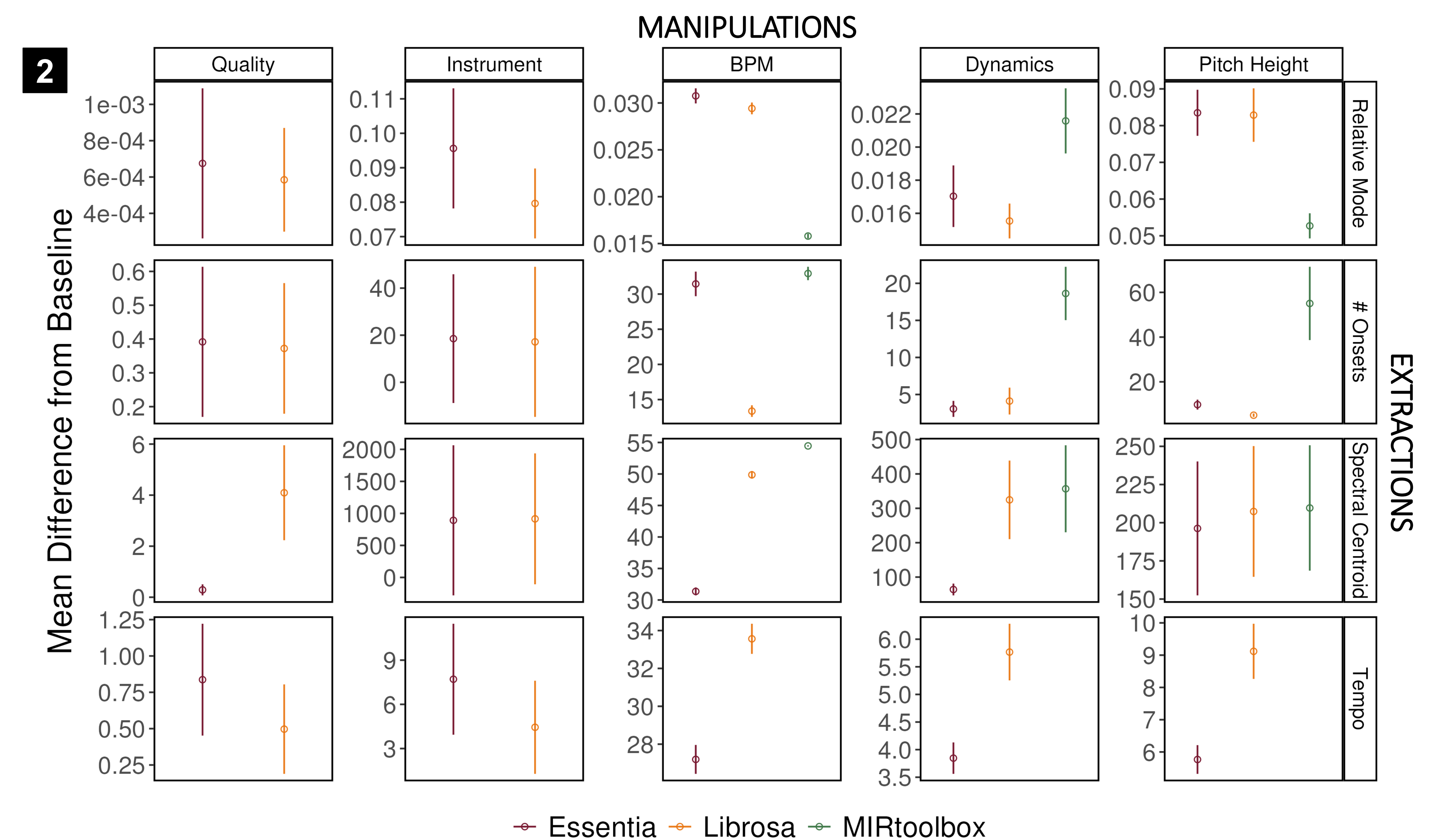
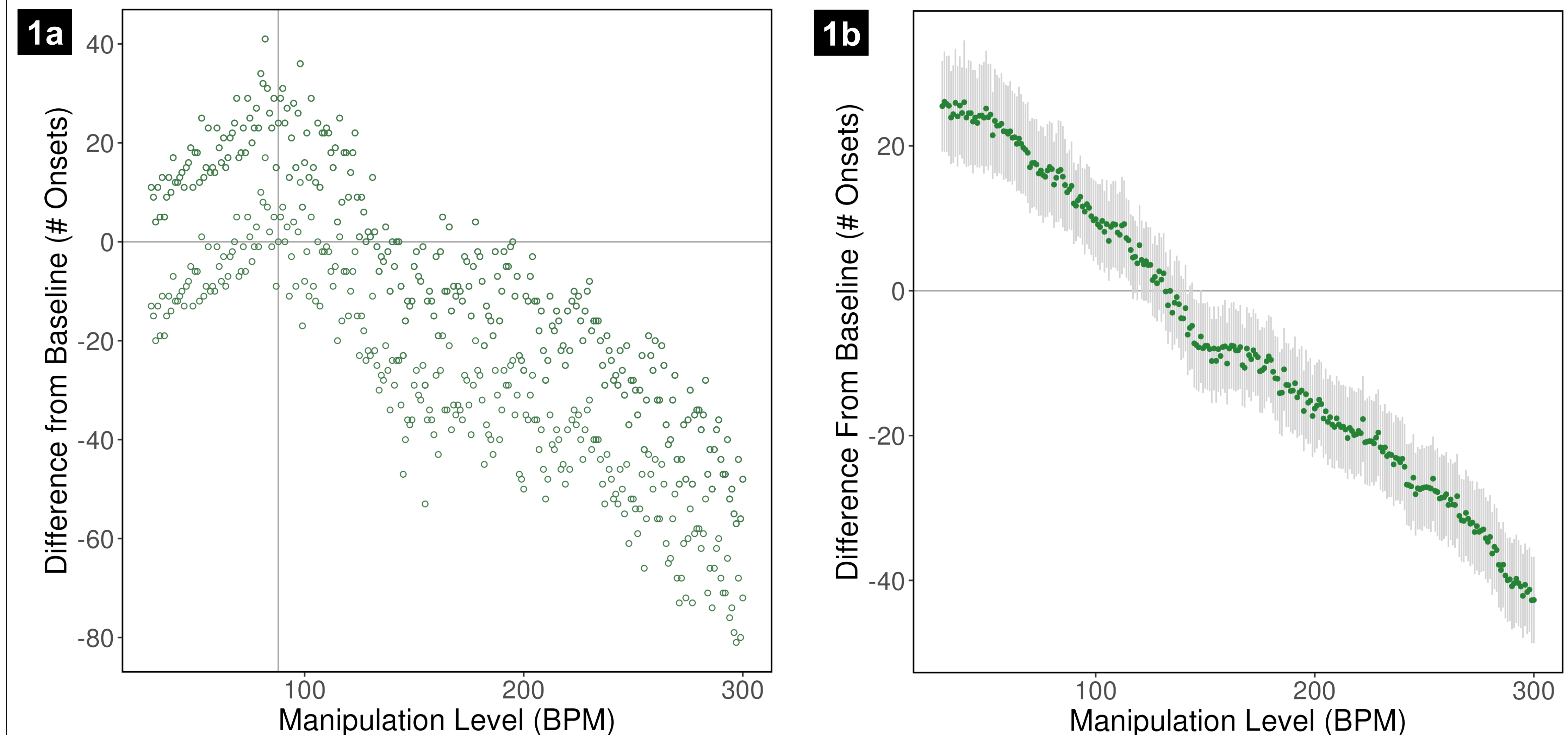


Figure 1a) One example of manipulations across one piece (Bach's WTC Prelude in F# Minor). Shows the difference in onsets as extracted by MIRtoolbox for manipulations of tempo between 30 and 300 BPM. The baseline tempo value for this piece is 88 BPM.
Figure 1b) A summary of manipulations for all 72 pieces in the corpus. Points indicate the mean of 72 pieces for that manipulation level and the gray lines indicate a 95% confidence interval.
Figure 2) Mean and 95% confidence interval for differences from baseline across 3 tools and 4 features for 5 different manipulations